CAP High School Prize Exam

5 April 2007

Marker's Information Sheet

The following information will be used to inform competitors and schools of the exam results, to determine eligibility for some subsequent competitions, and for statistical purposes. Only the marking code, to be assigned by the local examination committee, will be used to identify papers for marking.

Marking Code:

Sponsored by:

Canadian Association of Physicists
Canadian Chemistry and Physics Olympiads

Canadian Association of Physicists 2007 Prize Exam

This is a three-hour exam. National ranking and prizes will be based on a student's performance on sections A, B, and C of the exam. Performance on the questions in parts A and B will be used to determine whose written work in part C will be marked for prize consideration by the CAP Exam National Committee. Part A consists of twenty multiple-choice questions; part B consists of five questions that require a graphic solution. The problems in part C can also require graphing.

Non-programmable calculators may be used. The multiple-choice questions must be **on the answer card/sheet** provided. Solutions to the three long problems must be written on **three separate** sheets, as they will be marked by people in different parts of Canada.

Data

Speed of light $c = 3.00\text{x}10^8$ m/s
Gravitational constant $G = 6.67\text{x}10^{-11}$ N·m^2/kg^2
Acceleration due to gravity $g = 9.80$ m/s^2
Density of fresh water $\rho = 1.00\text{x}10^3$ kg/m^3
The normal atmospheric pressure $P_0 = 1.01\text{x}10^5$Pa
Fundamental charge $e = 1.60\text{x}10^{-19}$ C
Mass of electron $m_e = 9.11\text{x}10^{-31}$ kg
Mass of proton $m_p = 1.67\text{x}\ 10^{-27}$ kg
Planck's constant $h = 6.63\text{x}10^{-34}$ J·s
Coulomb's constant $1/(4\pi\varepsilon_0) = 8.99\text{x}10^9$ N·m^2/C^2
Boltzmann constant $k = 1.38\text{x}10^{-23}$ J/K

Part A: Multiple Choice

Question 1

A book is placed on a chair. Then a videocassette is placed on the book. The floor exerts a normal force

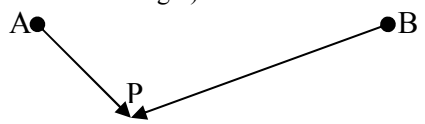
(a) on all three;
(b) only on the book;
(c) only on the chair;
(d) upwards on the chair and downwards on the book.

Solution

Answer **(c)** is correct because the floor is in contact only with the chair (the normal force is a force of reaction).

Question 2

The figure shows two point sources (A and B) of coherent mechanical waves of the same wavelength. Source B emits waves that are +π radians out of phase with the waves from source A. Source A is 3λ distant from P and source B is 5λ distant from P (λ is the wavelength).



The phase difference between the waves arriving at P from A and B is

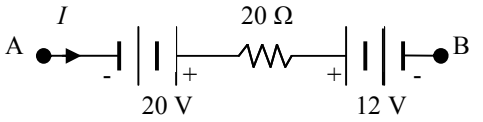
(a) 0 rad (b) π/2 rad (c) 2π rad (d) 3π rad (e) 4π rad

Solution

If two points are separated by the distance equal to the wavelength λ, the phase difference between oscillations of the wave in these points is equal to 2π rad. Without the initial phase difference between the waves from two sources, the paths difference in the point P would be 2λ, which is equivalent to the phase difference of 4π rad. As the wave, emitted from the point B, is π rad ahead relative to the phase of the wave from the point A, the resultant phase difference in the point P is equal to (4π – π) rad = 3π rad.

Answer **(d).**

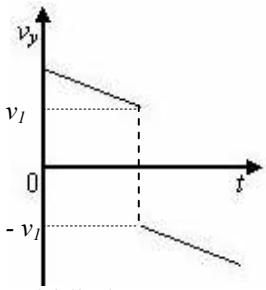
Question 3
What is the potential difference $V_B - V_A$ in the circuit segment below if the current $I = 1.5$ A?



(a) +22 V (b) –22 V (c) –38 V (d) +38 V (e) +2.0 V

Solution
$V_B - V_A$ = - 12 V + 20Ω x (- 1.5 A) + 20 V = - 22 V.
Answer **(b)** is correct.

Question 4
The graph on the right shows the velocity versus time graph for a ball. Which explanation best fits the motion of the ball as shown by the graph?



(a) The ball is falling, is caught, and is thrown down with greater velocity.
(b) The ball is rising, hits the ceiling, and falls down.
(c) The ball is falling, hits the floor, and bounces up.
(d) The ball is rising, is caught, and then is thrown down.

Solution
At the beginning, the motion must be decelerated. So, answers (a) and (c) are wrong. As in (d) it is said that the ball is caught, it means that the ball is stopped. Some time interval must be shown for the state of the ball in the hands of a person, who has caught it.
(b) is the correct answer.

Question 5
A light bulb A is rated at 60 W and a light bulb B is rated at 100 W. Both are designed to operate at 110 V. Which statement is correct?

(a) The 60 W bulb has a greater resistance and greater current than the 100 W bulb.
(b) The 60 W bulb has a greater resistance and smaller current than the 100 W bulb.
(c) The 60 W bulb has a smaller resistance and smaller current than the 100 W bulb.
(d) The 60 W bulb has a smaller resistance and greater current than the 100 W bulb.
(e) We need to know the resistivities of the filaments to answer this question.

Solution
All answers suppose the calculation of a ratio of resistances and currents for two bulbs. For resistances we use the formula for power P as: $P = V^2/R$; for currents the other formula is more convenient: P = IxV.
So, $R_{60} / R_{100} = 100 / 60 = 5/3$; and $I_{60} / I_{100} = 60 /100 = 3/5$.
Therefore, the bulb A has greater resistance and smaller current through it.
Answer **(b)** is correct.

Question 6
When a light ray travels between any two points, the path it takes is the one that

(a) covers the greatest distance;
(b) avoids travel in more than one medium;
(c) covers the least distance;
(d) takes the least time;
(e) is the mean between the longest and the shortest paths.

Answer **(d)** is correct according to the Fermat's principle.

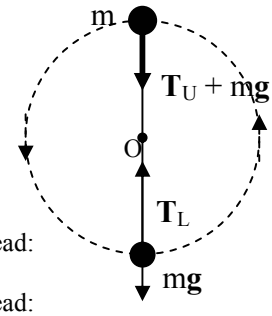
Question 7
A fountain sends water to a height of 10 meters. What must be the pressurization (above atmospheric pressure) of the water system driving the fountain? 1 ATM = 10^5 N/m^2.

(a) 10.8 ATM; (b) 9.80 ATM;
(c) 8.80 ATM; (d) 0.980 ATM.

Solution
$P_x + P_0 = \rho gh + P_0$, where P_0 is the atmospheric pressure; ρ is the density of water; g is acceleration to gravity; h is given to be 100 m. $P_x = \rho gh = 1.00\text{x}10^3 \text{ x } 9.80\text{x}10 / 10^5 = 0.980$ ATM.
(d) is the correct answer.

Question 8
A bead of mass *m*, attached to a string, is pushed and starts to rotate in the vertical plane. Only the gravitational force influences the rotation of the bead. The difference between the tension of the string when the bead is at the lowest point (T_L) and when the bead is at the upper point (T_U) is:

(a) $T_U - T_L = 2mg$;
(b) $T_U - T_L = 6mg$;
(c) $T_L - T_U = 2mg$;
(d) $T_L - T_U = 5mg$;
(e) $T_L - T_U = 6mg$.



Solution
For the lower position of the bead:
$T_L - mg = mv_L^2/R$ (1)
For the upper position of the bead:
$T_U + mg = mv_U^2/R$ (2)
The law of conservation of energy gives the following:
$mv_L^2/2 = mv_U^2/2 + 2mgR$ (3)
Subtracting equation (2) from (1) and taking into account (3), we can obtain that
$T_L - T_U = 2mg + 4mg = 6mg$
Answer **(e)**

Question 9

Two identical particles, each with a mass of 4.5 mg and charge of 30 nC, are moving directly toward each other with equal speeds of 4.0 m/s at the instant when the distance separating the two particles is equal to 25 cm. How far apart will they be from each other at the point of closest approach?

(a) 7.8 cm; (b) 9.8 cm; (c) 12 cm;
(d) 15 cm; (e) 20 cm.

Solution

According to the law of conservation of energy:

$$2\frac{mv^2}{2} + k\frac{q^2}{r_i} = k\frac{q^2}{x}$$, where $k = 9.0\text{x}10^9 \text{Nm}^2/\text{C}$.

Solving the equation for x, we obtain answer **(a)**

Question 10

A straight wire of length L carries a current I in the positive x direction in a region where the magnetic field is uniform and specified by $B_x = 3B$, $B_y = -2B$, and $B_z = B$, where B is a constant. What is the magnitude of the magnetic force on the wire?

(a) 1.0 ILB; (b) 2.2 ILB; (c) 3.6 ILB;
(d) 4.2 ILB; (e) 5.0 ILB

Solution

x-component of the magnetic field B_x does not act on the wire where current is directed in x-direction. A magnitude of a vector sum of y- and z-components of the magnetic field is equal to

$$B_\perp = \sqrt{(B)^2 + (-2B)^2} = \sqrt{5}B = 2.2B$$

The magnitude of the magnetic force on the wire is given by

$F = B_\perp IL = 2.2\ ILB$

Answer **(b)**

Question 11

A spaceship of mass m circles a planet of mass M in an orbit of radius R. How much energy is required to transfer the spaceship to a circular orbit of radius $3R$?

(a) $GmM/(2R)$; (b) $GmM/(3R)$; (c) $GmM/(4R)$;
(d) $GmM/(6R)$; (e) $3GmM/(4R)$

Solution

According to the law of conservation of energy:

$$\frac{mv_i^2}{2} - G\frac{Mm}{R} + W = \frac{mv_f^2}{2} - G\frac{Mm}{3R}$$

The centripetal force is the gravitational force:

$$\frac{mv_i^2}{R} = G\frac{Mm}{R^2}$$, therefore the total energy on each orbit is:

$$E_i = -\frac{1}{2}G\frac{Mm}{R}; \qquad E_f = -\frac{1}{2}G\frac{Mm}{3R}$$

$$W = -G\frac{Mm}{6R} - \left(-G\frac{Mm}{2R}\right) = G\frac{Mm}{3R}$$

Answer **(b)**

Question 12

Three simple pendulums with strings of different lengths and bobs of different masses are pulled out from the position of equilibrium to angles of θ_1, $\theta_2 = 2\theta_1$, and $\theta_3 = 3\theta_1$, respectively. All angles θ_1, θ_2 and θ_3 are very small. The bobs are then released and start oscillating freely. Which answer better matches the results of measurements of the frequencies of the three pendulums?

(a) $f_1 = 2f_2$; *and* $f_1 = 3f_3$;
(b) $f_3 = 3f_1$ *and* $f_2 = 2f_1$;
(c) $f_1 = f_2 = f_3$;
(d) Need to know the mass of each bob to find the relationship among the frequencies;
(e) Need to know the length of each pendulum to find the relationship among the frequencies.

Solution

For a simple pendulum oscillating at small angles of deflection, the frequency depends only on the length of the pendulum and on the value of g.

Therefore, the correct answer is **(e).**

Question 13

The electric potential inside a charged solid spherical conductor in equilibrium:

(a) is always zero;
(b) decreases from its value at the surface to a value of zero at the centre;
(c) is constant and is equal to its value at the surface;
(d) increases from its value at the surface to a higher value at the centre.

Solution

The potential difference between the surface of a conductor and any point inside the conductor is calculated according to: $\Delta V = -\int E dr = 0$, because the electric field inside the sphere is 0. Therefore, the potential remains constant inside the charged conductive sphere.

Correct answer is **(c).**

Question 14

For which process below will the internal energy of a system NOT change?

(a) An adiabatic expansion or compression of an ideal gas;
(b) An isothermal expansion or compression of an ideal gas;
(c) An isobaric expansion or compression of an ideal gas;
(d) The freezing of a quantity of a liquid at its melting point;
(e) The evaporation of a quantity of a liquid at its boiling point.

Solution

For an ideal gas, the internal energy is proportional to the temperature and remains constant for the isothermal process only, because in this process the temperature is constant.

Answers (d) and (e) are also applicable to processes with the constant temperature. However, the change of state is

accompanied by the change in the internal energy. In example (d) the internal energy decreases; in the example (e) the internal energy increases.
The correct answer is **(b).**

Question 15
A diver shines an underwater searchlight at the surface of a pond whose water has an index of refraction n =1.33. At what angle of incidence relative to the surface will the light be totally reflected?

(a) 41° (b) 47° (c) 49° (d) 51° (e) 58°

Solution
The angle of incidence of the initial ray on the surface is counted from the normal line to the surface. The critical angle of incidence α, is given by: $\sin \alpha = 1/n$; and $\alpha = 49^o$. The angle of the incident beam relative to the surface is given by $90^o - \alpha = 41^o$.
The correct answer is **(a).**

Question 16
According to the Bohr model of the atom, an electron can undergo a transition from one orbit that is closer to the nucleus to another which is farther from the nucleus, by absorbing a photon whose energy E depends on its frequency f as $E = hf$, where h is Planck's constant. Energy of 13.6 eV is needed to ionize a hydrogen atom by ejecting an electron from the lowest energy level. What is the longest wavelength of the photon that can eject the electron from the lowest energy level of the atom?

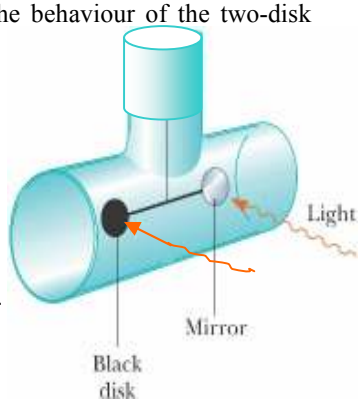
(a) 40 nm; (b) 60 nm; (c) 70 nm; (d) 80 nm; (e)90 nm.

Solution
To answer the problem, we need to convert electron volts in joules and use the relationship:
$E = hc/\lambda \rightarrow \lambda = hc/E = 90$ nm
Answer **(e).**

Question 17
One reason why we know that magnetic fields are not the same as electric fields is because the force exerted on a charge $+q$, is:
(a) in opposite directions in electric and magnetic fields if the charge is moving;
(b) parallel to the magnetic field and perpendicular to the electric field if the charge is moving;
(c) parallel to the electric field and perpendicular to the magnetic field if the charge is moving;
(d) zero in the electric field and nonzero in the magnetic field if the charge is not moving;

The correct answer is **(c).**

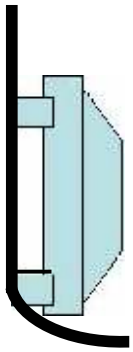
Question 18
In an experiment to prove the existence of the light pressure, two vertically oriented disks are attached as shown to the ends of a horizontal beam in an evacuated tube. The horizontal beam is suspended at its central point by a vertical wire. The surfaces of the disks are simultaneously illuminated by a parallel beam of light of high intensity. Which explanation best fits the behaviour of the two-disk system?



(a) The horizontal beam is displaced from its equilibrium position due to the equal light pressure on both disks.
(b) The horizontal beam rotates around the vertical wire with the mirror moving in the direction of light propagation and the black disk moving in the opposite direction.
(c) The horizontal beam rotates around the vertical wire with the black disk moving in the direction of light propagation and the mirror moving in the opposite direction.
(d) The experiment cannot show the existence of light pressure because light has no mass.

Solution
The correct answer is **(b)** , because the mirror experiences a greater force from the light beam changing its momentum by $2p_0$ due to the reflection (the analogue of the perfectly elastic collision), while the black disk experiences only one half of this force absorbing the incident momentum totally (the analogue of the perfectly inelastic collision).

Question 19
You are shown a photo of a car driven on a vertical inside wall of a huge cylinder with a radius of 50 m. The coefficient of static friction between the car tires and the cylinder is $\mu_s = 0.8$. The minimum speed, at which the car can be driven like that, is:



(a) 20 km/h; (b) 70 km/h; (c) 90 km/h;
(d) 120 km/h.

Solution
Force of friction f_S is directed upwards and relates to the normal force N as: $f_S = N \mu_S$. Minimum velocity v corresponds to $f_S = mg$.
Normal force has to provide centripetal force:
$N = mar = mv^2/r$
$\mu_S mv^2/r = mg$
$v^2 = g\,r/\mu_S = 9.8 \cdot 50/0.8\ m^2/s^2$
$v = 25 m/s = 90 km/h$
The correct answer is **(c).**

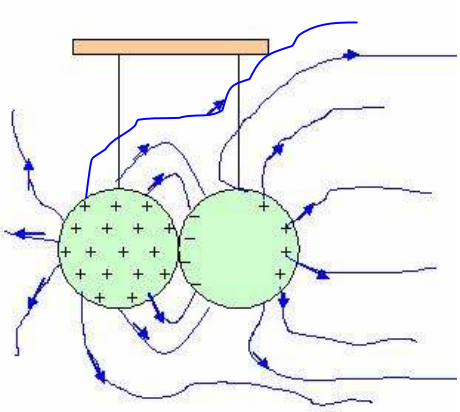
Question 20
An emf may be induced in
(a) a piece of linear wire when it moves in a uniform static magnetic field;

(b) a closed loop of wire moving with a fixed orientation at a constant velocity in a non-uniform static magnetic field;
(c) a closed loop of wire moving with a fixed orientation and accelerating in a uniform static magnetic field;
(d) the cases described in (a) and (b) only;
(e) the cases described in (b) and (c) only.

The correct answer is **(d)**

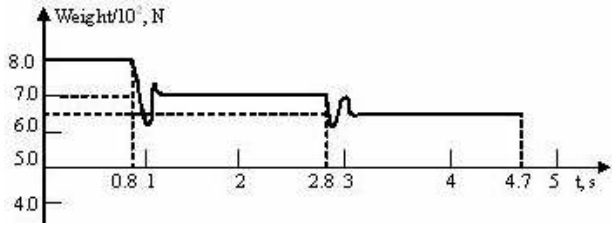
Part B: Questions that require graphical solutions
Question 21
Two identical dielectric balls supported by insulating threads hang side by side, touching each other. The two balls are initially electrically neutral.
Sketch the position of the balls on their threads after one of the balls is positively charged and the other stays neutral.
Sketch the electric field lines near the balls.

Solution
The material of both balls is an insulator. Therefore, charges cannot travel inside the balls. However, the electric field of the charged ball induces the polarization of molecules in the neutral ball and the neutral ball may be treated as a dipole. However, the electric field of the dipole is weaker than the field between the charged ball and the negative "end" of the neutral ball. Therefore, the field lines everywhere have the horizontal component from the charged ball to the neutral ball. The position of the balls and their threads does not change relatively to the initial one, as the balls attract each other.

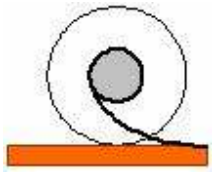


Question 22
A 71-kg man stands on a spring scale in an elevator. Starting from rest, the elevator ascends, attaining its maximum speed of 1.2 m/s in 0.80 s. It travels with this constant speed for the next 2.0 seconds. The elevator then undergoes a uniform acceleration downwards (in the negative y direction) for 1.9 s and comes to rest.
Draw a diagram for the reading of the scale versus time during the motion of the elevator.

Solution
During the first 0.80 seconds, the elevator was moving upwards with constant acceleration; the scale showed the constant weight = 71 x (9.8 + 1.2/0.80) N = 8.0×10^2N.
During the next 2.0 seconds, the man traveled in an inertial frame of reference; and the scale showed the weight = 71 x 9.8 N = 7.0×10^2N.
During the last 1.9 seconds, the scale showed the weight = 71 x (9.8 – 1.2/1.9) N = 6.5×10^2 N.
At the moment the acceleration of the elevator changes, the spring of the scale starts to vibrate. We have no data about the elastic properties of the spring and should assume that vibrations damp very fast, so that a tiny time interval is needed to change the value of the spring's expansion or compression. Therefore, the real process cannot be shown as three perfect steps on the graph.



Question 23
A spool of thread on a horizontal surface may roll without slipping if someone pulls it by the free end of the thread. The spool may roll toward or away from the person who pulls the free end of the thread depending on the direction of the applied force.
Sketch a side-view diagram of the spool of thread on the horizontal surface which shows the direction of the force applied to the free end of the thread in such a way that this force does not cause the rolling of the spool in either direction. The force belongs to the plane of the drawing.



Solution
Rolling is instantaneous rotation about the contact point P. The gravitational and normal forces produce no torque about this point. In Figure 23.1, the force $\boldsymbol{F_0}$, applied at the critical angle θ_C,, exerts zero torque about the spool's contact point with the ground and so will not make the spool roll.

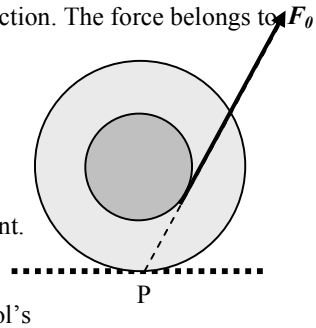


Fig. 23.1

Students are expected to draw only one diagram like in Fig. 23.1.
From the right-angle triangle, shown in the supplementary Figure 23.2, observe that
$\theta_C, = 90^o - \phi = 90^o - (90^o - \gamma) = \gamma$
Thus, $\cos \theta_C, = \cos \gamma = r / R$.

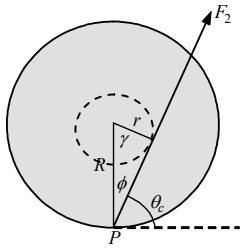
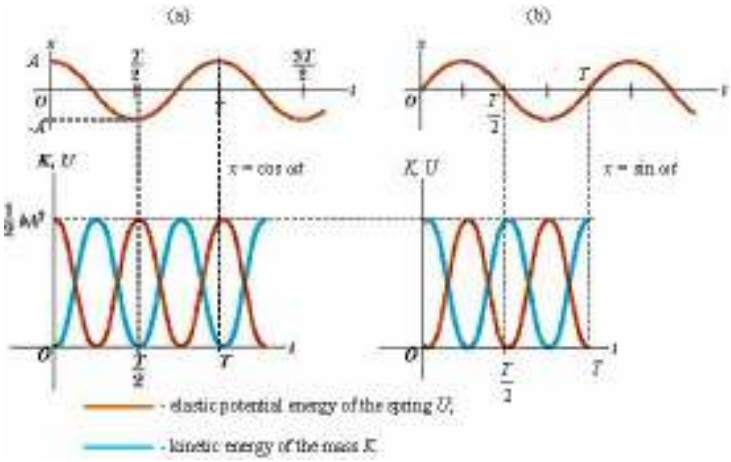


Fig. 23.2

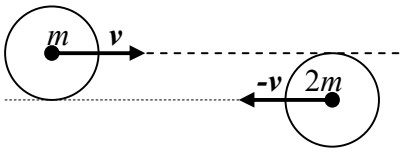
The direction of $\boldsymbol{F_0}$ depends on the radius of the layer of threads on the spool (r) and the external radius of the spool (R).

Question 24
Consider the motion of a mass attached to a spring.
Sketch the following two graphs one below the other using the same scale for the time axis (in units of period):
1) Displacement of the mass versus time;
2) Kinetic energy of the mass versus time.

Solution
Before sketching the graphs, we should write the formulae for (a) the displacement vs. time; (b) the elastic potential energy of the spring vs. time and the kinetic energy of the mass vs. time. Let us assume that the mass starts to move from the position with maximum displacement from the position of equilibrium (graph (a)).
(a) $x = A \cos \omega t$;
(b) $U = ½ kx^2 = ½ m\omega^2A^2\cos^2(\omega t)$;
$K = E – U = ½ kA^2 – ½ kx^2 = ½m\omega^2A^2\sin^2(\omega t)$.
Both diagrams (a) or (b) are correct. Students are expected to draw only the blue graphs for kinetic energy without graphs for potential energy.



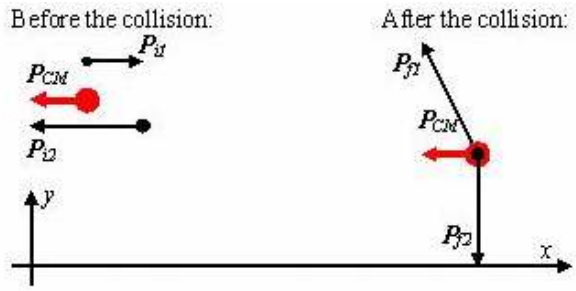
Question 25
A system consists of two ideal spheres with equal diameters that move toward each other as shown.



The spheres undergo a glancing collision.
Sketch a vector diagram for the linear momentum of each of the spheres and for the centre of mass of the system before and after the perfectly elastic collision of the spheres.

Solution
There are only internal forces between two spheres, and therefore the linear momentum of the system is conserved. This means that at any instant of time the linear momentum of the center of mass must be same. The collision is not head-on; therefore, the spheres scatter after the collision. The diagram for the momenta before the collision must show the relationship between two initial momenta ($p_2 = 2p_1$); the magnitude and the direction of the total momentum of the system. The system of coordinate is essential.

The diagram after the collision may vary if students have not enough time to solve the problem with two laws of conservation. However, it must show (1) that y-components of the two momenta are equal in magnitude and oppositely directed; (2) that the sum of x-components is equal to the initial total momentum of the system; and (3) that the less massive particle experiences the greater change in direction of motion and has the greater angle of scattering back.



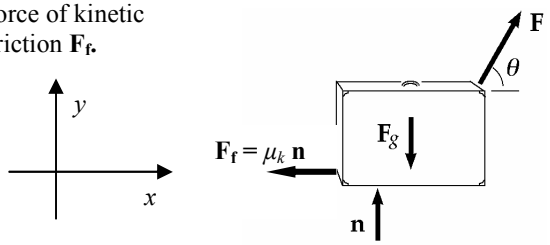
Part C: Problems
Problem 1
A student, working in a laboratory, has to remove a heavy box with new equipment from its initial position on the floor to the opposite wall of the room, which is *d* metres away. She may choose: either to pick up the box, to carry it in her hands across the room, and then to drop it on the floor in the new place; or to attach a rope to one of the corners of the box and pull the box along the floor to the new place. The mass *M* of the box and its contents is printed on the package. Acceleration due to gravity *g*, the coefficients of static and kinetic friction μ_s and μ_k for the box material and the floor, and other constants are available to the student in a handbook. After performing some calculations, the student notes that $\mu_k < \mu_s < 1$ and chooses the second method.

1) What physical quantities may the student compare to choose the proper method of moving the heavy box?
2) What parameter related to a quantity above shows an advantage for one of the two methods? Estimate and compare this parameter for the two methods.
3) Draw a free-body diagram (i.e. a vector diagram) for the chosen method of pulling the box with the attached rope along the horizontal floor.
4) Dragging the box may be easier or harder. Express the condition that makes dragging easiest in terms of physics. Identify the parameter responsible for this condition using your free-body diagram.
5) Using the parameter found, give a numerical condition for the easiest way of moving the box by the student.

Solution
1) The student could compare the maximum force, required to apply to the box, and the total work to be done on the box. We can assume that the mass of the load was rather great and therefore the student compared the maximum force.

2) The maximum force to remove the box shows the advantage of the second method. The maximum force to apply to pick up the box in a vertical direction is its gravitational force, or F_V = Mg. Dragging the box at a constant speed with the help of a rope, inclined to the horizontal with the angle θ, the student applies the force $F_{incl} = \mu_k Mg/(\cos\theta + \mu_k \sin\theta)$ (1), where $\mu_k < 1$. Varying θ from 0 to 90°, the student may apply the force from $\mu_k Mg < Mg$ to Mg, which corresponds the vertical lifting. Therefore, any angle $\theta < 90°$ gives an advantage in the applied force compared to the method of lifting the box vertically upward.

3) The free-body diagram includes the applied force **F**, the gravitational force $\mathbf{F_g}$, the normal force **n**, and the force of kinetic friction $\mathbf{F_f}$.



4) In terms of Physics, "easier" means with the minimum applied force F. From the free-body diagram:

n + Fsin $\theta = Mg$

F cos$\theta = \mu_k$ n = μ_k (Mg - Fsin θ)

Solving this system, we obtain the formula (1):

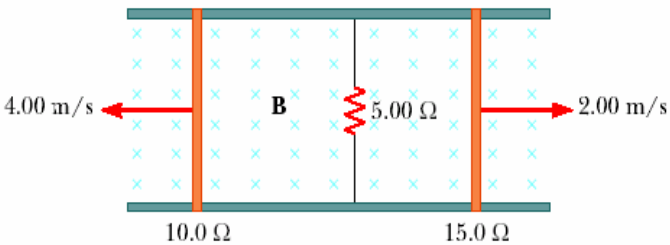
F = $\mu_k Mg/(\cos\theta + \mu_k \sin\theta)$. To find F_{min}, we may find the maximum for the expression $\cos\theta + \mu_k \sin\theta$. Thus, the parameter responsible for "easier" or "harder" removing the box is the angle θ.

5) The first derivative of the value $\cos\theta + \mu_k \sin\theta$ with respect to the angle θ is

$-\sin\theta + \mu_k \cos\theta = 0$, which gives the condition for the minimum force F as follows: $\tan\theta = \mu_k < 1$. We can conclude that depending on the coefficient of friction the angle $\theta < 45°$.

Problem 2

Two parallel rails with negligible resistance are 10.0 cm apart and are connected by a 5.00-Ω resistor. The circuit also contains two metal rods having resistances of 10.0 Ω and 15.0 Ω sliding along the rails as shown in the figure. The rods are pulled away from the resistor at constant speeds of 4.00 m/s and 2.00 m/s, respectively. A uniform magnetic field of magnitude 0.0100 T is applied perpendicular to the plane of the rails.



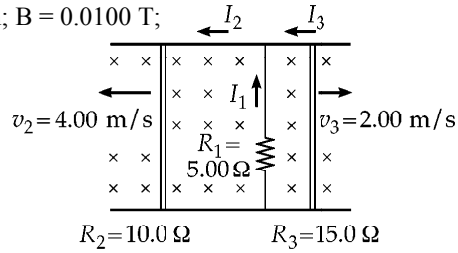
1) Determine the direction and the value of the current in the 5.00-Ω resistor.
2) Find forces, applied to the 10.0-Ω and the 15.0-Ω rods.
3) What is the force, applied to the segment of the circuit with the 5.00-Ω resistor?
4) If the segment with the 5.00-Ω rod could slide along the rails and was released at a given time, how would this segment move (qualitatively)?

Solution

Name the currents as shown in the diagram:

d = 10.0 cm; B = 0.0100 T;

1) Left loop:



$$+Bdv_2 - I_2R_2 - I_1R_1 = 0$$

Right loop: $+Bdv_3 - I_3R_3 + I_1R_1 = 0$

At the junction: $I_2 = I_1 + I_3$

Then, $Bdv_2 - I_1R_2 - I_3R_2 - I_1R_1 = 0$

$$I_3 = \frac{Bdv_3}{R_3} + \frac{I_1R_1}{R_3}.$$

So, $Bdv_2 - I_1(R_1 + R_2) - \frac{Bdv_3R_2}{R_3} - \frac{I_1R_1R_2}{R_3} = 0$

$$I_1 = Bd\left(\frac{v_2R_3 - v_3R_2}{R_1R_2 + R_1R_3 + R_2R_3}\right) = 145\ \mu\text{A (upwards)}$$

2) $I_2 = \dfrac{Bdv_2 - I_1R_1}{R_2} = 328\mu A$; $F_2 = I_2dB = 3.28\text{x}10^{-7}$N;

Magnetic force of the same magnitude opposes the external force and is directed to the right.

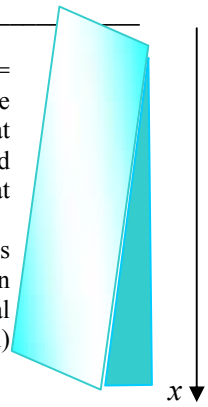
$$I_3 = \frac{Bdv_3 + I_1R_1}{R_3} = 182\mu A;\ F_3 = I_3dB = 1.82\text{x}10^{-7}\text{N}$$

3) $F_1 = I_1dB = 1.45\text{x}10^{-7}$N; F_1 is directed to the left (right-hand rule). After the bar with the 5.00-Ω resistor is released, it starts to accelerate to the left until the instant of time when its speed reaches 4 m/s. Since this moment, the current I_1 changes its direction to the opposite one and the bar starts to decelerate, then stops and starts moving to the right. Therefore, the bar oscillates, but the position of equilibrium is not stationary. The equilibrium position moves to the left, and oscillation of the bar is not harmonic.

Problem 3

A soap film with an index of refraction n = 1.33 is contained within a rectangular wire frame. The frame is held vertically so that the film drains downwards and approximates the shape of a wedge with flat faces near the top.

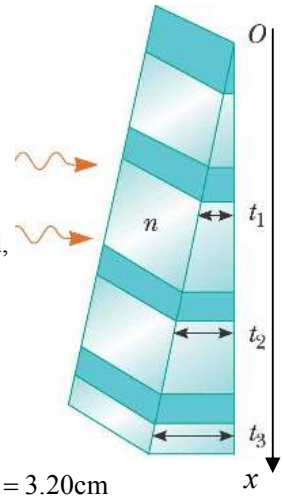
The thickness of the film at the very top is essentially zero. The film is viewed in reflected white light with near-normal incidence, and the first green (λ = 525 nm)



interference band is observed 4.00 cm from the top edge of the film.

1) Locate the first violet (λ = 420 nm) interference band.
2) Determine the film thickness at the positions of the first violet and the first green bands.
3) What is the wedge angle of the film?

Solution



1) Bright bands are observed when $2nt = \left(m + \frac{1}{2}\right)\lambda$.

The first bright band ($m = 0$) corresponds to $nt = \frac{\lambda}{4}$. Let a subscript 1 identify the green band, and the subscript 2 identify the violet band.

Since $\frac{x_1}{x_2} = \frac{t_1}{t_2}$ (see the figure), we have

$$x_2 = x_1 \frac{t_2}{t_1} = x_1 \frac{\lambda_2}{\lambda_1} = 4.00\text{cm}\ \frac{420}{525} = 3.20\text{cm}$$

2) For the violet band:

$$t_2 = \frac{\lambda_2}{4n} = \frac{420\text{nm}}{4(1.33)} = 78.9\text{nm}$$

For the green band:

$$t_1 = \frac{\lambda_1}{4n} = \frac{525\text{nm}}{4(1.33)} = 98.7\text{nm}$$

3)

$$\theta \approx \tan\theta = \frac{t_1}{x_1} = \frac{98.7\text{nm}}{4.00\text{cm}} = 2.47 \times 10^{-6}\ \text{rad}$$